



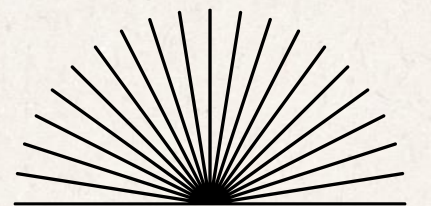
10/24/2025

Bank Term Deposit Subscription Prediction

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Brown University

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GitHub:
[https://github.com/Judyluo1217/
/Bank-Term-Deposit-
Subscription-Prediction.git](https://github.com/Judyluo1217/Bank-Term-Deposit-Subscription-Prediction.git)



Agenda

03	Intro
04	EDA - Duration vs. Campaign
05	EDA - Pre-Call Clues
06	EDA - Multicollinearity
07	Split Method
08	Missing Values
09	Preprocessing - Strategic Feature Removal
10	Preprocessing - Handling Data Types & Final Output

01 Problem

Predict client subscription ('yes'/'no')

02 Importance

Improve marketing efficiency, target right clients

03 Task Type

Binary Classification

04 Data Source

UCI Machine Learning Repository (Bank Marketing Dataset).

05 Data Collection

Portuguese bank phone campaigns

06 Difficulties

Missing Data (labeled 'unknown')

Intro

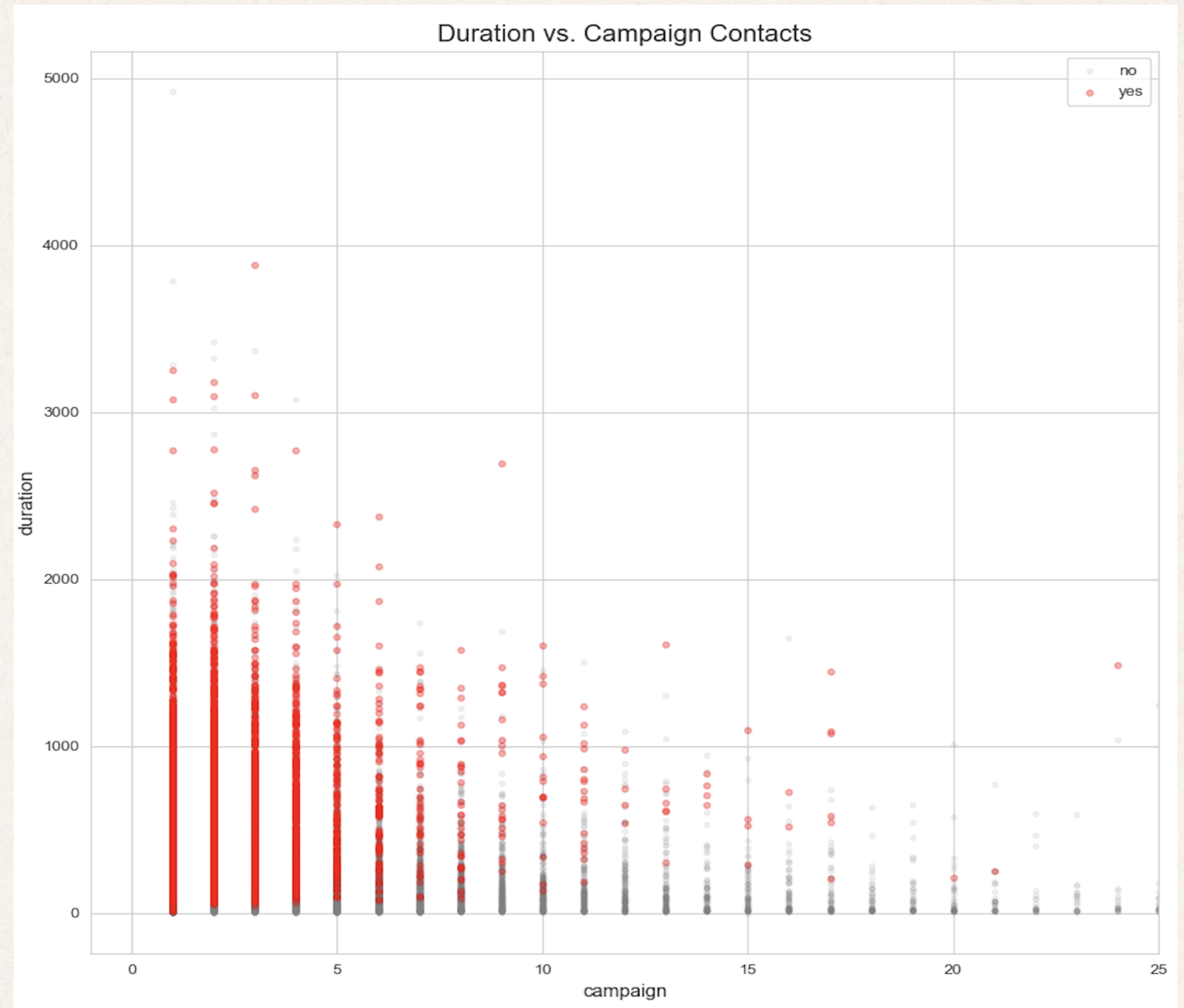
Introduce the project.
Provide a quick background..

Duration vs. Campaign (continuous)

04/10

Quality > quantity

- High duration = 'yes', Low duration = 'no'
- Most 'yes' in first few contacts
- Quality > Quantity
- PROBLEM on using duration: data leakage

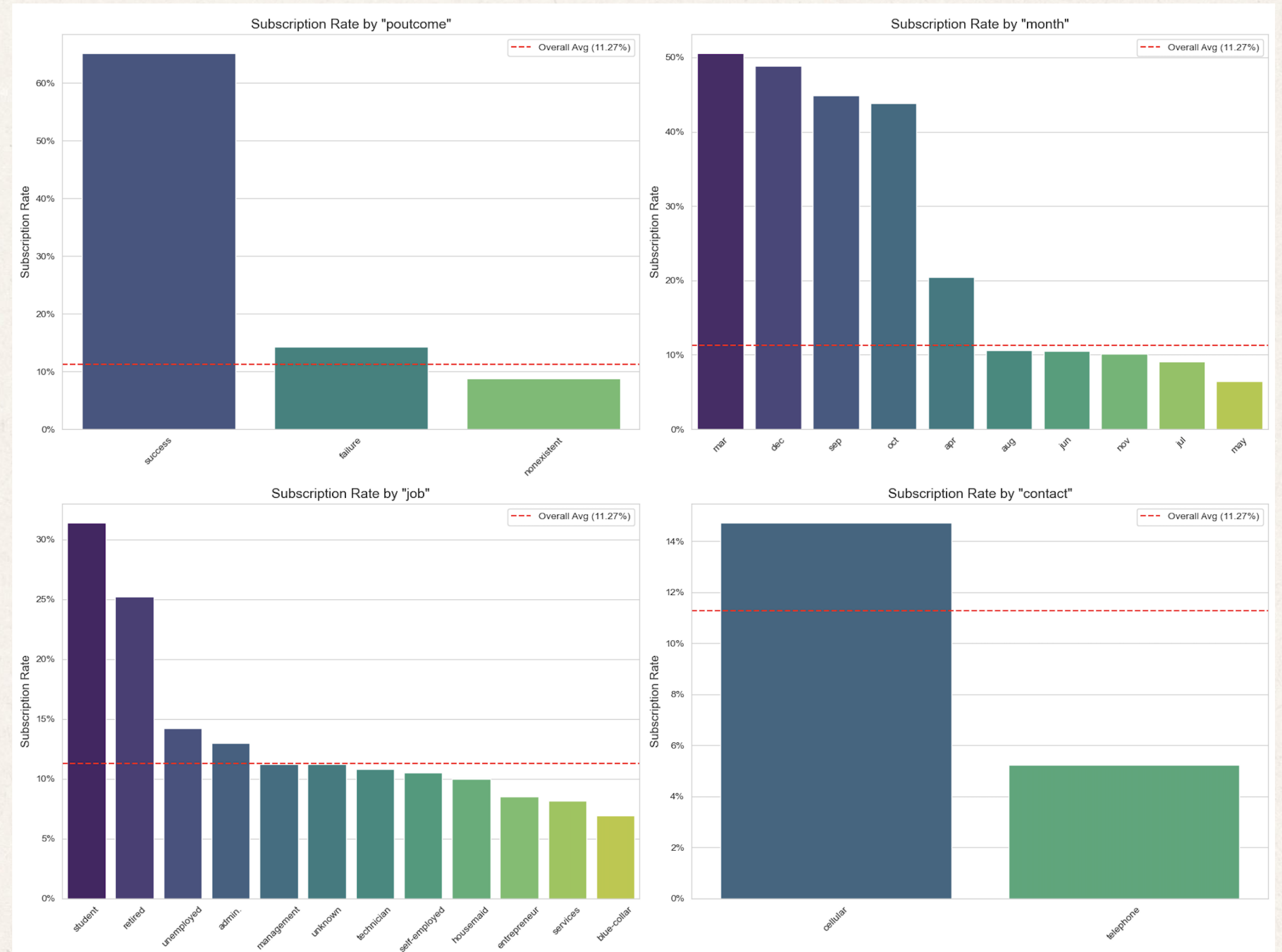


Pre-Call Clues (categorical)

05/10

Identifying High-Potential Clients & Timing

- Previous Outcome: 'success' (>60% rate)
- Job: 'student', 'retired'
- Month: Mar, Sep, Oct, Dec
- Contact: 'cellular'

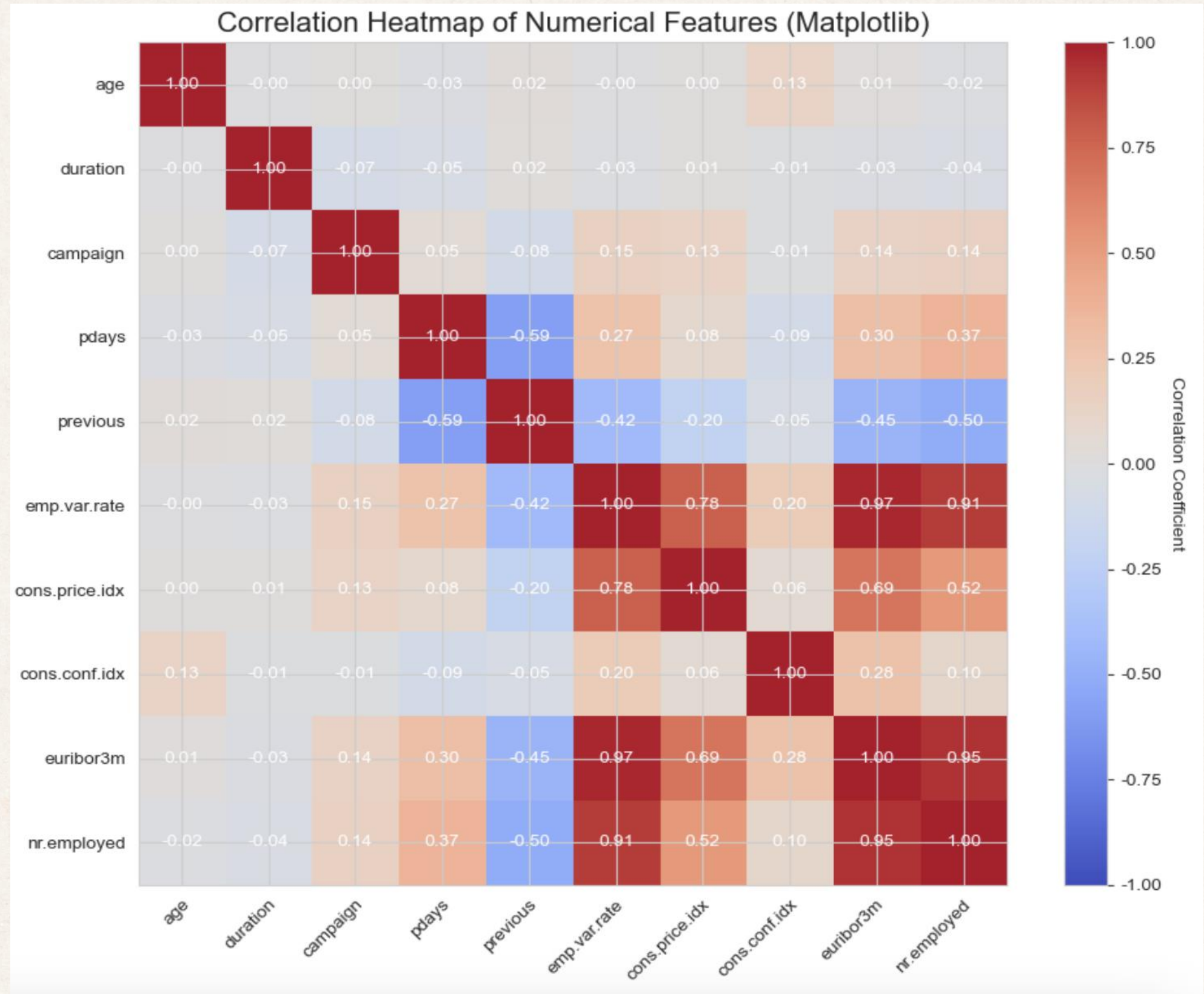


Multicollinearity

06/10

Addressing High Feature Correlation

- **Problem:** High correlation (>0.9) in socio-economic features
- **Features:** emp.var.rate (employment variation rate), euribor3m (euribor 3 months rate), nr.employed (number of employees)
- **Impact:** A known risk for linear models, but may be handled by more complex models (e.g., NNs, Trees)



Split Method

07/10

Target Variable

Distinct minority class (11.3% 'yes')

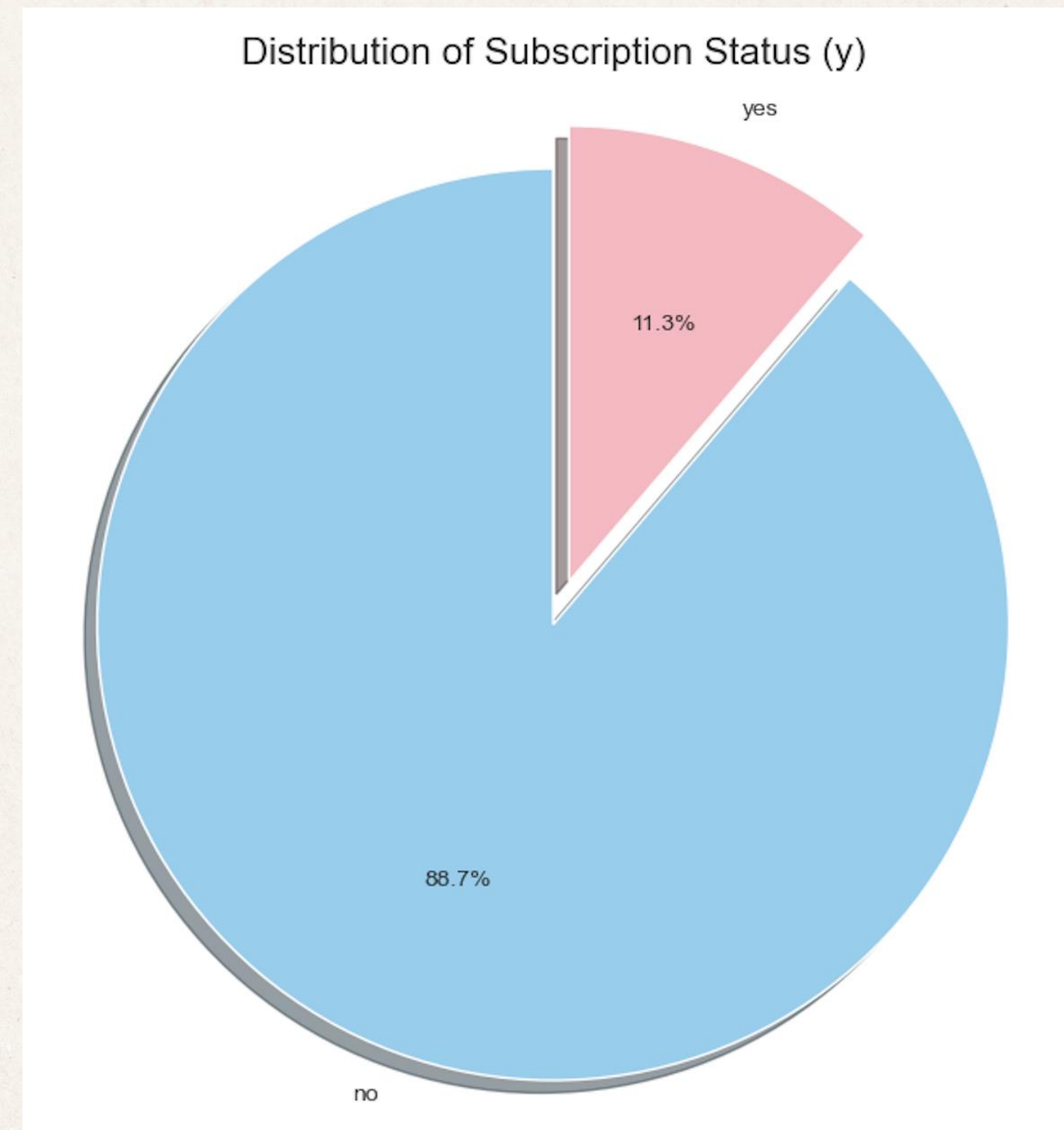
Train/Test Split (80/20)

Stratified Split (stratify=y)

Why? With ~45k samples

Cross-Validation Setup (on 80% Training Data)

10-Fold Stratified Cross-Validation (StratifiedKFold)



Missing Value

- Definition:

Missing values = 'unknown'

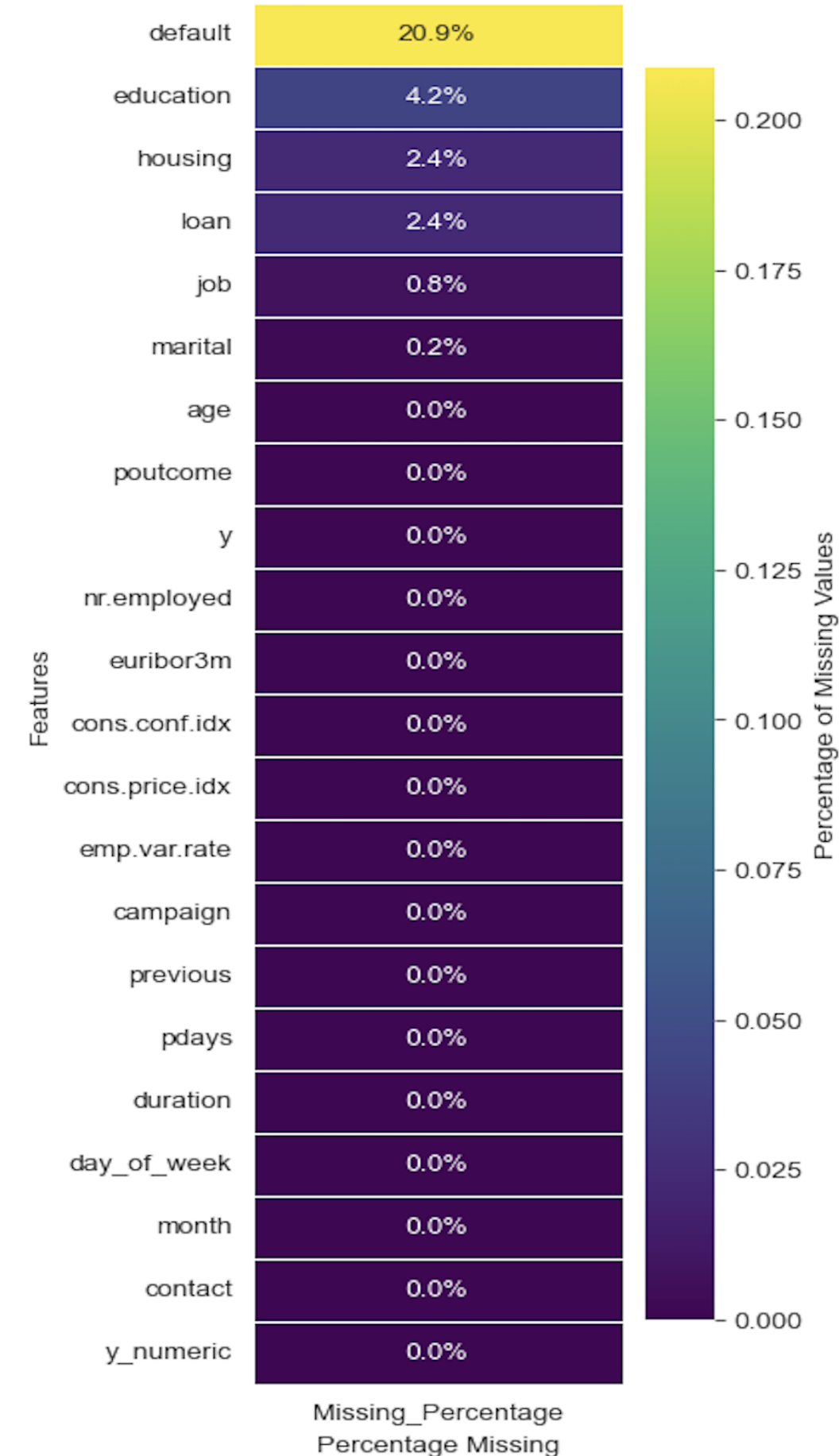
- Top Features:

'default' (20.9%), 'education' (4.2%)

- Overall Impact:

~21% of rows contain at least one 'unknown'

Comprehensive Missing Value Analysis (Vertical)



Preprocessing

09/10

Feature Selection Rationale

GOAL: Realistic pre-call prediction
Removed duration (last contact duration): duration (unavailable pre-call)
Multicollinearity: Keep all three (emp.var.rate, euribor3m, nr.employed) Might offer unique value
Result: 20 features → 19 features

Preprocessing

Transforming Features

10/10

Approach:

Used ColumnTransformer: mixed data → numerical matrix

Numeric Pipeline:

StandardScaler(): mean 0 and std 1

Ordinal Pipeline :

OrdinalEncoder(): ordered categories → integers

Nominal Pipeline:

OneHotEncoder(): new binary (0/1) columns

Final Output:

19 features → 42 features

Thank you

CONTACT

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